

# Optimal Fiscal Policy without Heterogeneity

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## 1 Deterministic Economy

- Judd (1985): Capital Taxation without Public Debt

## 2 Stochastic Economy

- Zhu (1992): Capital and Labor Taxation: a Tractable Case
- CCK (1994): Capital and Labor Taxation: a General Case

## 3 Economy with Market Power and Price Stickiness

- SGU (2004): Labor Taxation with Monetary Authority

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# Decentralized Environment

- Two agents: capitalists  $c$  and non-capitalists (workers, which is hand-to-mouth)  $n$ , sharing the same preference,

$$u[c(t)] = \frac{[c(t)]^{1-\sigma}}{1-\sigma}.$$

- Labor supply:  $L^c = 0$ ,  $L^n = L = 1$ , both inelastically.
- Production function:  $F(k) = F(k, 1)$ , satisfying standard assumptions. and  $f(k)$  is the product net of depreciation (with rate  $\delta > 0$ ).
- Government:
  - $\tau(t)$ , capital income tax rate.
  - $G$ , government spending.
  - $T^c$  and  $T^n$ , lump-sum transfer / tax on capitalists and workers, respectively.

# Decentralized Environment

- Capitalists' problem: given  $w$ ,  $r$ , and  $\tau(t) \rightarrow \tau^1$ ,

$$\max_{c(t), k(t)} \int_0^{\infty} e^{-\rho t} u[c(t)] dt$$

s.t.

$$c + \dot{k} = wL^c + (r - \delta)(1 - \tau)k + T^c,$$

where  $k(0) = k_0$  is given.

- Euler equation:

$$\frac{\dot{c}}{c} = \frac{(r - \delta)(1 - \tau) - \rho}{\sigma}.$$

- Transversality condition:

$$\lim_{t \rightarrow \infty} e^{-\rho t} u'[c(t)] = 0.$$

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<sup>1</sup>This strong assumption will ensure the existence of a steady state and convergence to that steady state. We will discuss it later.

# Competitive Equilibrium

- FOCs on prices:

$$\begin{aligned}r &= f'(k) + \delta, \\w &= f(k) - kf'(k).\end{aligned}$$

- Thus, the system of a competitive equilibrium can be written as<sup>2</sup>

$$\begin{aligned}\dot{C} &= -C \frac{\rho - (1 - \tau)f'(k)}{\sigma}, \\ \dot{k} &= (1 - \tau)kf'(k) - C,\end{aligned}$$

where  $C$  is total consumption by capitalists per worker.

- The steady-state capital stock satisfies

$$f'(k^*) = \frac{\rho}{1 - \tau}.$$

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<sup>2</sup>We only concern the case that all capital income tax receipts are redistributed uniformly among the workers.

# The Ramsey Problem

- Let  $\bar{r}(t)$  be the rate of return net of both taxes and depreciation realized by capitalists at  $t$ , that is,  $\bar{r}(t) \equiv [r(t) - \delta][1 - \tau(t)]$ .
- Suppose the Ramsey planner concerns only with the welfare of the workers determines a tax policy for all future time and that such revenues must cover a constant stream of government consumption,  $G$ , as well as lump-sum transfers to the workers.

# The Ramsey Problem

- Then, the Ramsey problem becomes

$$\max_{\bar{r}(t)} \int_0^{\infty} e^{-\rho t} u[c(t)] dt$$

s.t.

$$c + G + \bar{r}k = f(k), \quad (\text{RC})$$

$$\dot{C} = -C \frac{\rho - \bar{r}}{\sigma}, \quad (\text{IC})$$

$$\dot{k} = \bar{r}k - C, \quad (\text{IC})$$

$$\bar{r} \geq 0, \quad (\text{tax bound})$$

$$\lim_{t \rightarrow \infty} k(t) \in \mathbb{R}_+, \quad \lim_{t \rightarrow \infty} C(t) \in \mathbb{R}_+.$$

# Saddle Manifold and Shooting Method

- Since the two-dimensional system  $(\dot{k}, \dot{C})$  is saddle, we can use shooting method to solve the transition paths of  $k(t)$  and  $C(t)$ . Specifically, given  $k(0) = k_0$ , choosing  $C(0)$  such that  $(k, C) \rightarrow (k^*, C^*)$ .
- Then, resolve the after-tax interest rate path  $\bar{r}(t) = \Gamma(k, C)$  by FOCs. Eventually we obtain the tax rate  $\tau(t) = 1 - \bar{r}(t)/f'[k(t)]$ .
- In fact, the tax bound will never bind. We omit this constraint in the following discussion.

# The Recursive Form

- Straub and Werning (2020) studied the recursive form of Judd (1985) with discrete time:

$$V(k, C_-) = \max_{(k', C) \in Z^*} U(c) + \beta V(k', C)$$

s.t.

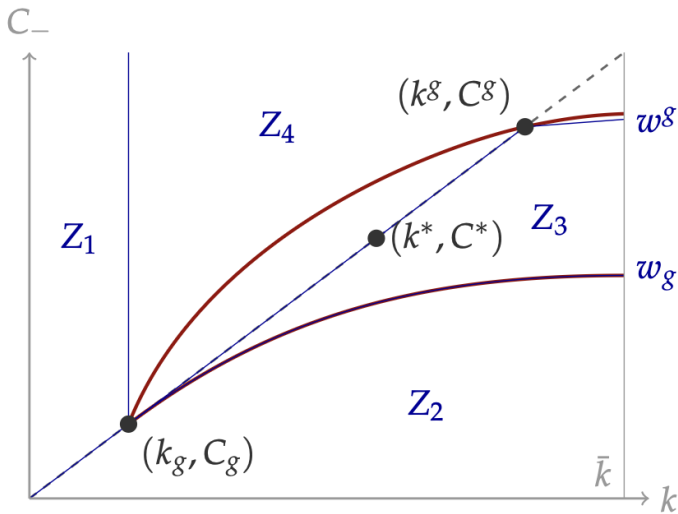
$$c + C + k' + g = f(k) + (1 - \delta)k, \quad (\text{RC})$$

$$\beta U'(C)(C + k') = U'(C_-)k, \quad (\text{IC})$$

$$c, C, k' \geq 0.$$

- At  $t = 0$ , capital  $k_0$  is given, and the planner maximizes  $V(k_0, C_{-1})$  w.r.t.  $C_{-1}$ .

# Characterizing the Feasible Set



# Characterizing the Feasible Set

- To solve the Bellman equation we must first compute the feasible set  $Z^*$ . We restrict the range of capital to a closed interval  $[k_{\min}, k_{\max}]$  with  $k_{\min} \geq k_g$ . This leads us to seek a subset  $Z^{*k_{\min}}$  of the feasible set  $Z^*$ .
- Start with  $Z_{(0)}^*$  defined by  $C_- = \frac{1-\beta}{\beta}k$  and  $k \in [k_{\min}, k_{\max}]$ . This set is self-generating and thus  $Z_{(0)}^* \subset Z^{*k_{\min}}$ .
- Define an operator that finds all points  $(k, C_-)$  for which one can find the allocation satisfying the constraints of the Bellman equation and  $(k, C) \in Z_{(0)}^*$ . This gives a set  $Z_{(1)}^* \supset Z_{(0)}^*$ .
- Given tolerance, we iterate on this procedure and obtain  $Z_{(0)}^*, Z_{(1)}^*, \dots$ , till  $Z_{(n)}^* \approx Z^*$ .

# Value Function Iteration

- We then solve the Bellman equation by value function iteration.
- Our procedure uses a grid that is defined on a transformation of  $(k, C_-)$  that maps  $Z^*$  into a rectangle. We linearly interpolate between grid points.

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- The representative agent's preference:

$$U = \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t u(c_t, l_t),$$

and leisure  $l_t = 1 - n_t$ .

- Neoclassical production function:  $y_t = F(k_t, n_t; z_t)$ , where  $\{z_t\}$  is a exogenous stochastic process.
- Government spending  $\{g_t\}$  is also a given stochastic process. So in period  $t$ , the state of the economy is represented by history  $s^t \equiv (s_0, s_1, \dots, s_t)$ , where  $s_t = (g_t, z_t)$ .
- Resource constraint:

$$c_t + g_t + k_{t+1} = F(k_t, n_t; z_t) + (1 - \delta)k_t.$$

# Decentralized Environment

- Boundary on capital income tax  $\tau_t$ :

$$\int p_t(1 - \tau_t)r_t ds_t \geq 0,$$

where  $p_t$  is the PDF of one-period-forward contingent claims.

- Agent's budget constraint:

$$c_t + k_{t+1} \leq (1 - \theta_t)w_t n_t + \bar{R}_t k_t + b_t - \int p_{t+1} b_{t+1} dz_{t+1},$$

where  $\theta_t$  is the labor income tax rate,  $\bar{R}_t = (1 - \delta) + (1 - \tau_t)r_t$  is the gross capital income tax rate.

- Then we can characterize the individual's problem by FOCs and TVCs.
- TVCs become

$$\begin{aligned}\lim_{t \rightarrow \infty} \beta^t \mathbb{E}_0 u_{c,t} b_t &= 0, \\ \lim_{t \rightarrow \infty} \beta \mathbb{E}_0 u_{c,t} \bar{R}_t k_t &= 0.\end{aligned}$$

- MRS between consumption and leisure:

$$\frac{u_{n,t}}{u_{c,t}} = (1 - \theta_t)w_t.$$

- Euler equation:

$$u_c(c_t, 1 - n_t) = \beta \mathbb{E}_t u_{c,t+1} \bar{R}_{t+1}.$$

- Law of motion of  $p_t$ :

$$p_{t+1}(s_{t+1}|s^t) = \beta \frac{\pi_{t+1}}{\pi_t} \frac{u_{c,t+1}}{u_{c,t}},$$

if  $\pi_t > 0$ . Here,  $\pi_t$  is the given PDF of  $s^t$ .

- The solution firm's problem is standard:

$$\begin{aligned}r_t &= F_k(k_t, n_t; z_t), \\w_t &= F_n(k_t, n_t; z_t).\end{aligned}$$

- In this situation, the allocation  $\{c_t, n_t, k_{t+1}\}$  is called a competitive allocation that can be implemented by the policy  $\{\tau_t, \theta_t, b_{t+1}\}$ .

- One can prove that in this economy, the levels of public debt payment in future periods and *ex post* capital income tax rates are indeterminate.
- Besides, the market values of public debt issue, the labor income tax rate, and the *ex ante* expected capital income tax rates that are consistent with certain competitive allocation are uniquely determined.
- Notice that this indeterminacy result is different from Barro (1979), the Ricardian equivalence theorem. Because the economy in this model is stochastic in which the government can issue contingent claim bonds.

- The market value of public debt issue:

$$B_t := \int p_{t+1} b_{t+1} ds_{t+1}.$$

- The *ex ante* expected capital income tax rate:

$$\bar{\tau}_t := \frac{\mathbb{E}_{t-1} u_{c,t} \tau_t r_t}{\mathbb{E}_{t-1} u_{c,t} r_t}.$$

- Using the law of motion of  $p_t$  to rewrite this equation, we have

$$\bar{\tau}_t = \frac{\int p_t \tau_t r_t k_t ds_t}{\int p_t r_t k_t ds_t},$$

that is, the *ex ante* capital income tax rate can be interpreted as the certainty equivalent capital income tax rate.

# The Ramsey Problem

- To make the analysis simple and tractable, the following assumptions are made:

(i) log-separable utility

$$u(c, 1 - n) = \ln c + \gamma \ln(1 - n).$$

(ii) Cobb-Douglas production

$$y = F(k, n; z) = k^\alpha n^{1-\alpha} m(z).$$

(iii) full depreciation:  $\delta = 1$ .

- Let  $\Phi$  be the multiplier on implementability constraint.
- Under these assumptions, the Ramsey allocation is characterized by the following conditions.

# The Ramsey Allocation

- FOCs:

$$\begin{aligned}\frac{\gamma c_t}{1 - n_t} &= \frac{1 - n_t}{\Phi + 1 - n_t} (1 - \alpha) k_t^n n_t^{-\alpha} m(z_t), \quad t > 0, \\ \frac{\gamma c_0}{1 - n_0} \left( 1 + \frac{\Phi b_0}{c_0} \right)^{-1} &= \frac{1 - n_0}{\Phi + 1 - n_0} (1 - \alpha) k_0^n n_0^{-\alpha} m(z_0), \\ c_y^{-1} &= \alpha \beta \mathbb{E}_t c_t^{-1} k_t^{\alpha-1} n_t^{1-\alpha} m(z_{t+1}), \quad t > 0, \\ c_0^{-1} \left( 1 + \frac{\Phi b_0}{c_0} \right) &= \alpha \beta \mathbb{E}_0 c_i^{-1} k_i^{\alpha-1} n_i^{1-\alpha} m(z_1); \end{aligned}$$

# The Ramsey Allocation

- Technology constraint:

$$g_t + c_t + k_{t+1} = k_t^\alpha n_t^{1-\alpha} m(z_t), \quad t \geq 0;$$

- Implementability constraint:

$$c_0^{-1} b_0 = \sum_{t=0}^{\infty} \mathbb{E}_0 \beta^t \left( 1 - \frac{\gamma n_t}{1 - n_t} \right);$$

- TVC:

$$\lim_{t \rightarrow \infty} \mathbb{E}_0 \beta^t \frac{k_{t+1}}{c_t} = 0.$$

# Optimal Fiscal Policies

- Then, we derive the determination optimal policies. Firstly, denote that  $\omega_t \equiv g_t/y_t$  and  $a_t \equiv c_t/y_t$ .
- The optimal labor income rates:

$$\theta_t = \frac{\Phi}{\Phi + 1 - n(\Phi, a_t)}.$$

- The optimal *ex ante* expected capital income tax rates:

$$\tau_t = \begin{cases} 1, & t = 0 \\ 0, & t \geq 1 \end{cases}.$$

- The market values of the optimal public debt issue:

$$B_t = \left[ -(1 - \omega_t) + a_t \left[ 1 + \mathbb{E}_t \sum_{j=1}^{\infty} \beta^j \left( 1 - \frac{\gamma n_{t+j}}{1 - n_{t+j}} \right) \right] \right] y_t.$$

- In the Ramsey equilibrium, all the optimal policies are functions of  $\omega_t$  and the constant  $\Phi$ . And  $\Phi$  itself is a function of initial values  $b_0$ ,  $\omega_0$ , and the initial distribution of  $\omega_t$ 's.
- Therefore, the limiting behavior of the Ramsey equilibrium depends on these initial values as well as the initial distribution of  $\omega_t$ 's. Thus, the Ramsey equilibrium is completely determined by the sequence of government expenditures to output ratios.
- For simplicity, we assume that there is no public debt outstanding in the initial period, i.e.,  $b_0 = 0$ .

- In this case, we have

$$a_0 = \left[ \mathbb{E}_0 \sum_{j=0}^{\infty} \frac{(\alpha\beta)^j}{\prod_{i=0}^j (1 - \omega_i)} \right],$$

and

$$n_0 = 1 - \frac{(1 - \Phi)\gamma a_0 + \sqrt{4(1 - \alpha)a_0\Phi + [\gamma a_0(1 + \Phi)]^2}}{2(1 - \alpha + \gamma a_0)},$$

and

$$k_1/y_0 = 1 - \omega_0 - a_0.$$

- The value of  $\Phi$  is determined by the condition

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left[ 1 - \frac{\gamma a(\Phi, a_t)}{1 - n(\Phi, a_t)} \right] = 0,$$

which can be solved by numerical methods such as Newton-Raphson.

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# Decentralized Environment

- History  $s^t \equiv (s_0, s_1, \dots, s_t)$  with unconditional probability  $\mu_t = \mu(s^t)$ . The initial realization  $s_0$  is given.
- Resource constraint<sup>3</sup>:

$$c_t + g_t + k_{t+1} = F(k_t, l_t, s_t) + (1 - \delta)k_t.$$

- The representative agent's preference follows standard assumptions:

$$\bar{U} = \sum_{t, s^t} \beta^t \mu_t U(c_t, l_t).$$

- Fiscal Instruments:
  - Labor income tax with rate  $\tau_t$ .
  - Capital income tax with rate  $\theta_t$ .
  - Public debt which has a one-period maturity and a state-contingent return, with the quantity of debt issued  $b_t$  and its payoff rate  $R_{b,t}$ .

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<sup>3</sup>Here, we write  $c_t = c(s^t)$ ,  $g_t = g(s^t)$ , and so do other state-contingent variables.

- Agent's budget constraint:

$$c_t + k_{t+1} + b_{t+1} \leq (1 - \tau_t)w_t l_t + R_{b,t} b_t + R_{k,t} k_t,$$

where  $R_{k,t} = 1 + (1 - \theta_t)(r_t - \delta)$  denotes the after-tax gross return rate.

- Competitive prices:

$$\begin{aligned} r_t &= F_k(k_t, l_t, s_t) \\ w_t &= F_l(k_t, l_t, s_t). \end{aligned}$$

- Let  $x_t = (c_t, l_t, k_{t+1}, b_{t+1})$  denote an allocation for consumers at  $s^t$ , and let  $x = [x_t]$  denote an allocation for all  $s^t$ .

- Government budget constraint:

$$R_{b,t}b_t + g_t = b_{t+1} + \theta_t(r_t - \delta)k_t + \tau_t w_t l_t.$$

- Let  $\pi_t = (\tau_t, \theta_t, R_{b,t})$  denote the fiscal policy at  $s^t$ , and let  $\pi = [\pi_t]$  denote the policy for all  $s^t$ . The initial stock of debt  $b_0$  and the initial stock of capital  $k_0$  are given.
- Note that the after-tax return rate are constrained to be non-negative, and the quantity of public debt are bounded above and below by some arbitrarily large constants.

# The Ramsey Problem

- In a full-commitment Ramsey problem, allocation rules are sequences of functions  $x(\pi) = [x(s^t|\pi)]$  that construct the map  $\pi \rightarrow x(\pi)$ . Prices rules are sequences of functions  $w(\pi) = [w(s^t|\pi)]$  and  $r(\pi) = [r(s^t|\pi)]$  that construct similar maps.
- A Ramsey equilibrium is a policy  $\pi$ , an allocation rule  $x(\cdot)$ , and price rules  $w(\cdot)$  and  $r(\cdot)$  such that
  - (i) the policy  $\pi$  maximizes  $\bar{U}$  *conditional on it*, subject to the government budget constraint with allocations and prices given by  $x(\pi)$ ,  $w(\pi)$ , and  $r(\pi)$ ;
  - (ii)  $\forall \pi'$ , the allocation  $x(\pi')$  maximizes  $\bar{U}$  subject to the agent's budget constraint evaluated at the policy  $\pi'$  and the prices  $w(\pi')$  and  $r(\pi')$ ;
  - (iii)  $\forall \pi'$ , the prices satisfy

$$w_{t|\pi'} = F_l[k_{t|\pi'}, l_{t|\pi'}, s_t],$$

$$r_{t|\pi'} = F_k[k_{t|\pi'}, l_{t|\pi'}, s_t].$$

# The Ramsey Problem

- It is well known that in a Ramsey equilibrium, the planner has an incentive to set  $\theta_0$  as large as possible. Thus, we adopt the convention that  $\theta_0$  and  $R_{b,0}$  are fixed.
- One can prove that the allocations  $(c_t, l_t, k_{t+1})$  in a Ramsey equilibrium solve the Ramsey allocation problem

$$\max \bar{U} = \sum_{t,s^t} \beta^t \mu_t U(c_t, l_t)$$

s.t.

$$c_t + g_t + k_{t+1} = F(k_t, l_t, s_t) + (1 - \delta)k_t, \quad (\text{RC})$$

$$\sum_{t,s^t} \beta^t \mu_t (U_{c,t} c_t + U_{l,t} l_t) = U_{c,0} [R_{k,0} k_0 + R_{b,0} b_0]. \quad (\text{IC})$$

# Optimal Fiscal Policies

- Now we derive the prices and policies in the Ramsey equilibrium.
- Substituting prices into agent's FOCs we obtain the intratemporal consumption-labor choice

$$-\frac{U_{l,t}}{U_{c,t}} = (1 - \tau_t)F_{l,t},$$

which determines the labor income tax rate, as well as two Euler equations

$$U_{c,t} = \sum_{s^{t+1}} \beta \mu_{t+1|t} U_{c,t+1} R_{b,t}$$
$$U_{c,t} = \sum_{s^{t+1}} \beta \mu_{t+1|t} U_{c,t+1} R_{k,t}$$

- The agent's budget constraint is

$$c_t + b_{t+1} + k_{t+1} = (1 - \tau_t)w_t l_t + R_{b,t} b_t + R_{k,t} k_t.$$

- Similar to Zhu (1992), one can prove the indeterminacy of capital tax rates. If  $R_b$  and  $\theta$  satisfy these equations, then so do any  $\bar{R}_b$  and  $\bar{\theta}$ , where

$$\begin{aligned}\sum_{s^{t+1}} \mu_{t+1|t} R_{b,t} &= \sum_{s^{t+1}} \mu_{t+1|t} \bar{R}_{b,t}, \\ \sum_{s^{t+1}} \mu_{t+1|t} U_{c,t+1} \theta_t (F_{k,t} - \delta) &= \sum_{s^{t+1}} \mu_{t+1|t} U_{c,t+1} \bar{\theta}_t (F_{k,t} - \delta), \\ \theta_t (F_{k,t} - \delta) k_t - R_{b,t} b_t &= \bar{\theta}_t (F_{k,t} - \delta) k_t - \bar{R}_{b,t} b_t.\end{aligned}$$

- If the government has either state-contingent capital taxes or state-contingent debt, it can support the Ramsey allocations.

# Optimal Fiscal Policies

- Then, we isolate certain fiscal variables that are uniquely determined.
- By the second equation, we derive the *ex ante* capital income tax rate as

$$\theta_t^e = \frac{\sum q_{t+1} \theta_t (F_{k,t} - \delta)}{\sum q_t (F_{k,t} - \delta)},$$

where  $q_{t+1} = \beta \mu_{t+1|t} U_{c,t+1} / U_{c,t}$  is the Arrow-Debreu price of a unit of consumption at state  $s^{t+1}$  in units of consumption at  $s^t$ .

- The certain equivalence of debt return rate  $\bar{r}_t$  satisfies

$$\sum_{s^{t+1}} q_{t+1} R_{b,t} = \sum_{s^{t+1}} q_{t+1} (1 + \bar{r}_t),$$

and by levying a state-contingent tax  $v_t$  on interest payments from public debt that satisfies  $R_{b,t+1} = 1 + \bar{r}_t (1 - v_t)$ , then we can derive the *ex ante* debt revenue tax rate as

$$\eta_t = \frac{\theta_t (F_{k,t} - \delta) k_t + v_t \bar{r}_t b_t}{(F_{k,t} - \delta) k_t + \bar{r}_t b_t}.$$

# Computational Procedure

- 1 Fix an initial value for  $\lambda$ , the multiplier on the implemenability constraint. Then solve for Ramsey allocations  $c(k, s)$ ,  $l(k, s)$  and  $k'(k, s)$ .
- 2 Solve for bond allocation rules  $b'(k, s)$ . Take  $\lambda$  as an optimal control. Iterate on  $\lambda$  to make  $R_0 b_0$  equal to the given value.
- 3 Solve for the tax policies  $\tau(\theta, s, \lambda)$ ,  $\theta^e(\theta, s, \lambda)$ , and  $\eta(\theta, s, \lambda)$ .

# Computational Procedure: Step 1

- Consider the Lagrangian of the Ramsey allocation problem,

$$\mathcal{L}(c_t, l_t, \lambda) = U(c, t, l_t) + \lambda(U_{c,t}c_t + U_{l,t}l_t).$$

- FOCs for  $\forall t \geq 1$ :

$$\begin{aligned} -W_{l,t}/W_{c,t} &= F_{l,t}, \\ W_{c,t} &= \sum_{s^{t+1}} \beta \mu_{t+1|t} W_{c,t+1} (1 - \delta + F_{k,t}). \end{aligned}$$

- FOCs for  $t = 0$ :

$$\begin{aligned} -\frac{W_{l,0} - \lambda[U_{cl,0}(R_{k,0}k_0 + R_{b,0}b_0) + U_{c,0}(1 - \theta_0)F_{kl,0}]}{W_{c,0} - \lambda U_{cc,0}(R_{k,0}k_0 + R_{b,0}b_0)} &= F_{l,0}, \\ W_{c,0} - \lambda U_{cc,0}(R_{k,0}k_0 + R_{b,0}b_0) &= \sum_{s^{t+1}} \beta \mu_{1|0} W_{c,1} (1 - \delta + F_{k,1}) = 0. \end{aligned}$$

# Computational Procedure: Step 1

For  $\forall t \geq 1$ :

- Use the resource constraint and the first FOC to express  $c(k, s)$  and  $l(k, s)$  in terms of  $k$ ,  $s$ , and  $k'(k, s)$ .
- Use the second FOC to determine the capital accumulation rule  $k'(k, s)$ .

For  $t = 0$ :

- Use the resource constraint and the third FOC to express  $c_0$  and  $l_0$  in terms of  $k_1$ .
- Use the forth FOC to determine  $k_1$ .

## Computational Procedure: Step 2

For  $\forall t \geq 1$ :

- Write the recursive form of the bond allocation:

$$\begin{aligned} U_c(k, s)b(k, s) = & \sum_{s'} \beta \mu(s'|s) [U_c(k', s')c(k', s') \\ & + U_l(k', s')l(k', s') + U_c(k', s')b(k', s') \\ & + U_c(k', s')k'(k', s')] - U_c(k, s)k', \end{aligned}$$

where  $k' = k'(k, s)$ .

- This form actually defines a contraction-mapping linear operator:  $b(k, s) \rightarrow b(k, s)$ . The stationary bond allocation rule exists and is the fixed point of that mapping.

For  $t = 0$ :

- Substitute  $c_0$  and  $l_0$  into  $U_c$  on the LHS of this equation and the stationary rules on the RHS to derive  $b_1(k_0, s_0)$ .

## Computational Procedure: Step 2

- Finally, we substitute  $b_1(k_0, s_0)$  together with the other period 0 allocation rules into the agent's budget constraint evaluated in period 0.
- Using the intratemporal condition and the after-tax wage rate and setting  $\theta_0$  to our initial rate (27.1% in this paper), we calculate a value for  $R_{b,0}b_0$ .
- We iterate on  $\lambda$  until the initial value for  $R_{b,0}b_0$  is the given value (0.20 in this paper), which is the steady-state value of government obligations in a deterministic economy.

## 1 Deterministic Economy

- Judd (1985): Capital Taxation without Public Debt

## 2 Stochastic Economy

- Zhu (1992): Capital and Labor Taxation: a Tractable Case
- CCK (1994): Capital and Labor Taxation: a General Case

## 3 Economy with Market Power and Price Stickiness

- SGU (2004): Labor Taxation with Monetary Authority

## 1 Deterministic Economy

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- Agents have the identical preference satisfying standard assumptions,

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t U(c_t, l_t).$$

- Two types of financial assets in each period  $t \geq 0$ : fiat money,  $M_t$ , and one-period nominal state-contingent assets  $D_{t+1}$ .
- Following MIU, we introduce money by assuming that it facilitates transactions. Specifically, consumption purchases are subject to a proportional transaction cost  $s(v_t)$  that depends on the agent's money-to-consumption ratio,  $v_t = P_t c_t / M_t$ .

# Decentralized Environment

- The final good  $c_t$  is assumed to be a composite good made of a continuum of intermediate differentiated goods  $z_t \tilde{l}_t$ , aggregated a la Dixit-Stiglitz.
- The labor market is perfect competitive.
- The intermediate input is of the form  $Y_t d(p_t)$ , where  $p_t = \tilde{P}_t / P_t$ . The demand function  $d(\cdot)$  is decreasing and satisfies  $d(1) = 1$  and  $d'(1) < -1^4$ .
- The production constraint on monopolists:

$$z_t \tilde{l}_t \geq Y_t d(p_t).$$

- We introduce sluggish price adjustment cost a la Rotemberg, with a parameter  $\theta$  measuring the degree of price stickiness:

$$\Theta_t = \frac{\theta}{2} \left( \frac{\tilde{P}_t}{\tilde{P}_{t-1}} - 1 \right)^2.$$

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<sup>4</sup>The restrictions on  $d(1)$  and  $d'(1)$  are necessary for the existence of a symmetric equilibrium.

# Decentralized Environment

- Agents' flow budget constraint in period  $t$ :

$$P_t c_t [1 + s(v_t)] + M_t + \mathbb{E}_t r_{t+1} D_{t+1} \\ \leq M_{t-1} + D_t + P_t \left[ \frac{\tilde{P}_t}{P_t} Y_t d \left( \frac{\tilde{P}_t}{P_t} \right) - w_t \tilde{h}_t - \Theta_t \right] + (1 - \tau_t) P_t w_t l_t.$$

- Borrowing constraint (non-Ponzi condition):

$$\lim_{j \rightarrow \infty} \mathbb{E}_t [q_{t+j+1} (D_{t+j+1} + M_{t+j})] \geq 0, \quad \forall t, s^t,$$

where  $q_t = r_1 r_2 \cdots r_t$  with  $q_0 \equiv 1$ .

- Then we can derive the FOCs of the agents' maximization problem.
- Denote  $\pi_t \equiv P_t / P_{t-1}$  as the gross price inflation rate,  $R_t = 1 / \mathbb{E}_t [r_{t+1}]$  as the gross risk-free nominal interest rate,  $\lambda_t / P_t$  as the multiplier on the flow budget constraint, and  $m c_t \lambda_t$  as the multiplier on the production constraint.

- The government faces a stream of public consumption, denoted by  $g_t$ ; that is exogenous, stochastic, and unproductive. These expenditures are financed by levying labor income taxes at the rate  $\tau_t$ ; by printing money, and by issuing one-period, risk-free (non-contingent), nominal obligations, which we denote by  $B_t$ .
- The government's sequential budget constraint:

$$M_t + B_t = M_{t-1} + R_{t-1}B_{t-1} + P_t g_t - \tau_t P_t w_t l_t, \quad \forall t \geq 0.$$

- The monetary-fiscal regime consists in the announcement of state-contingent plans for the nominal interest rate and the labor tax rate,  $\{R_t, \tau_t\}$ .

# Competitive Equilibrium

- We restrict attention to symmetric equilibria where all agents charge the same price for the good, i.e., identical. That is,  $\tilde{P}_t = P_t, \forall t \geq 0$ . It follows that  $\tilde{l}_t = l_t$ . Thus, the marginal revenue of monopolist is constant and equal to  $1 + 1/\eta > 0$ , where  $\eta \equiv d'(1) < -1$ .
- In equilibrium there is no borrowing or lending among the agents, that is,

$$D_t = R_{t-1}B_{t-1}, \quad \forall t, s^t.$$

- In equilibrium, it must be the case that the nominal interest rate is non-negative,

$$R_t \geq 1.$$

- A competitive equilibrium is a set of plans  $\{c_t, l_t, M_t, B_t, v_t, mc_t, \lambda_t, P_t, q_t, r_{t+1}, \Theta_t\}$  satisfying the following conditions: (to be cont'), given  $\{R_t, \tau_t\}$ ,  $\{z_t, g_t\}$ , and the initial condition  $R_{-1}B_{-1} + M_{-1} > 0$ .

# Competitive Equilibrium

- FOCs:

$$\begin{aligned}U_{c,t} &= \lambda_t[1 + s(v_t) + v_t s'(v_t)], \\ \frac{U_{l,t}}{U_{c,t}} &= \frac{(1 - \tau_t)z_t m c_t}{1 + s(v_t) + v_t s'(v_t)}, \\ v_t^2 s'(v_t) &= \frac{R_t - 1}{R_t}, \\ \lambda_t r_{t+1} &= \beta \lambda_{t+1} \frac{P_t}{P_{t+1}}.\end{aligned}$$

- Non-Ponzi condition:

$$\lim_{j \rightarrow \infty} \mathbb{E}_t[q_{t+j+1}(R_{t+j}B_{t+j} + M_{t+j})] = 0.$$

# Competitive Equilibrium

- NKPC:

$$\lambda_t \pi_t (\pi_t - 1) = \beta E_t [\lambda_{t+1} \pi_{t+1} (\pi_{t+1} - 1)] + \frac{\lambda_t \eta z_t l_t}{\theta} \left[ \frac{1 + \eta}{\eta} - mc_t \right].$$

- Government budget constraint:

$$M_t + B_t + \tau_t P_t z_t mc_t l_t = R_{t-1} B_{t-1} + M_{t-1} + P_t g_t.$$

- Resource constraint:

$$[1 + s(v_t)]c_t + g_t + \Theta_t = z_t l_t.$$

- and some definitions:  $R_t = 1/\mathbb{E}_t[r_{t+1}] \geq 1$ ,  $q_t = r_1 r_2 \cdots r_t$  with  $q_0 = 1$ ,  $v_t = P_t c_t / M_t$ , and  $\Theta_t = (\theta/2)(\pi_t - 1)^2$ .

# The Ramsey Problem

- It is difficult to apply primal approach:
  - In the model with sticky prices and non-state-contingent nominal public debt, the primal form of the competitive equilibrium cannot be stated in terms of a single time-zero implementability constraint, which is akin to Aiyagari, Marcet, Sargent, and Seppälä (2002).
- Why?
  - See our discussion.
- How?
  - Expressing the set of Ramsey constraints in terms of a sequence of intertemporal implementability constraints, rather than in terms of the sequence of TVC given as

$$\lim_{j \rightarrow \infty} \mathbb{E}_t \left[ \beta^j \frac{\lambda_{t+j+1}}{\pi_{t+j+1}} \left( \rho(v_{t+j}) b_{t+j} + \frac{c_{t+j}}{v_{t+j}} \right) \right] = 0,$$

where we denote  $\gamma(v_t) \equiv 1 + s(v_t) + v_t s'(v_t)$  and  $\rho(v_t) \equiv 1/[1 - v_t^2 s'(v_t)]$ .

# The Ramsey Problem

- One can prove that plans  $\{c_t, l_t, v_t, \pi_t, b_t, mc_t\}$  satisfying the following conditions: (to be cont'), and the boundary conditions on  $v_t$ ,  $v_t \geq v_{\min}$  and  $v_t^2 s'(v_t) < 1$ ,  $\forall t, s^t$ , and under all contingencies given  $(R_{-1}B_{-1} + M_{-1})/P_{-1}$ , are the same as those satisfying the definition of a competitive equilibrium.
- Resource constraint:

$$[1 + s(v_t)]c_t + g_t + \frac{\theta}{2}(\pi_t - 1)^2 = z_t l_t.$$

- NKPC:

$$\pi_t(\pi_t - 1) = \beta \mathbb{E}_t \left[ \frac{U_{c,t+1}}{U_{c,t}} \frac{\gamma(v_t)}{\gamma(v_{t+1})} \pi_{t+1}(\pi_{t+1} - 1) \right] + \frac{\eta z_t l_t}{\theta} \left[ \frac{1 + \eta}{\eta} - mc_t \right].$$

# The Ramsey Problem

- Sequential government budget constraint:

- $\forall t \geq 1$ :

$$\frac{c_t}{v_t} + b_t + \left[ mc_t z_t + \frac{U_{l,t} \gamma(v_t)}{U_{c,t}} \right] l_t = \frac{\rho(v_{t-1}) b_{t-1}}{\pi_t} + \frac{c_{t-1}}{v_{t-1} \pi_t} + g_t.$$

- $t = 0$ :

$$\frac{c_0}{v_0} + b_0 + \left[ mc_0 z_0 + \frac{U_{l,0} \gamma(v_0)}{U_{c,0}} \right] h_0 = \frac{R_{-1} B_{-1} + M_{t-1}}{P_{-1} \pi_0} + g_0.$$

- Sequence of intertemporal budget constraints:

$$\begin{aligned} \mathbb{E}_t \sum_{j=0}^{\infty} \beta^j \left[ U_{c,t+j} c_{t+j} + U_{l,t+j} l_{t+j} + z_{t+j} l_{t+j} (mc_{t+j} - 1) \frac{U_{c,t+j}}{\gamma(v_{t+j})} \right. \\ \left. + \frac{\theta}{2} (\pi_{t+j} - 1)^2 \frac{U_{c,t+j}}{\gamma(v_{t+j})} \right] = \frac{U_{c,t}}{\gamma(v_t)} \left[ \frac{c_{t-1}/v_{t-1} + \rho(v_{t-1}) b_{t-1}}{\pi_t} \right]. \end{aligned}$$

- For simplicity, we introduce the quantitative method in a flexible price economy.
- Then, the NKPC is eclipsed, and the resource constraint can be written as

$$[1 + \alpha s(v_t)]c_t + g_t = z_t l_t.$$

- Denote  $\psi_t$  and  $\xi$  as the multipliers on the resource constraint and implementability constraint (standard form), respectively.
- Denote  $\mathbb{I}(t = 0)$  as an indicator function that takes the value of one if  $t = 0$  and zero otherwise.

$$z_t h_t = [1 + \alpha s(v_t)] c_t + g_t, \quad (31)$$

$$E_0 \sum_{j=0}^{\infty} \beta^j \left\{ U_c(c_t, h_t) c_t \phi(v_t) + U_h(c_t, h_t) h_t + \frac{U_c(c_t, h_t)}{\gamma(v_t)} \frac{z_t h_t}{\eta} \right\} = \frac{U_c(c_0, h_0)}{\gamma(v_0)} d_0, \quad (32)$$

$$\xi U_c(c_t, h_t) c_t \phi'(v_t) - \xi U_c(c_t, h_t) \frac{\gamma'(v_t)}{\gamma^2(v_t)} \left[ \frac{z_t h_t}{\eta} - I(t=0) d_0 \right] = \psi_t \alpha s'(v_t) c_t, \quad (33)$$

$$\begin{aligned} U_c(c_t h_t) [1 + \xi \phi(v_t)] + \xi \left[ U_{cc}(c_t, h_t) \left( c_t \phi(v_t) - I(t=0) \frac{d_0}{\gamma(v_t)} \right) \right. \\ \left. + U_{hc}(c_t, h_t) h_t + U_{cc}(c_t, h_t) \frac{z_t h_t}{\gamma(v_t) \eta} \right] = \psi_t [1 + \alpha s(v_t)], \end{aligned} \quad (34)$$

$$\begin{aligned} -z_t \psi_t = U_h(c_t, h_t) + \xi \left[ U_{ch}(c_t, h_t) \left( c_t \phi(v_t) - I(t=0) \frac{d_0}{\gamma(v_t)} \right) \right. \\ \left. + U_h(c_t, h_t) + h_t U_{hh}(c_t, h_t) + U_{ch}(c_t, h_t) \frac{z_t h_t}{\gamma(v_t) \eta} + U_c(c_t, h_t) \frac{z_t}{\gamma(v_t) \eta} \right] \end{aligned} \quad (35)$$

# Quantitative Solution: Step 1-3

- **Step 1:** Assume that  $g_t$  and  $z_t$  follow independent 2-state symmetric Markov processes<sup>5</sup>. Then the possible states of the economy are described by the  $4 \times 1$  state vector  $S$ . The  $4 \times 4$  transition matrix of  $S$  is denoted by  $\Phi$ .
- **Step 2:** Choose an initial state  $s_0 \equiv (g_0, z_0)$ .
- **Step 3:** Choose a positive value for  $\xi$ .

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<sup>5</sup>The generalization of the method to  $N$ -state Markov processes is trivial.

## Step 4:

- For  $t > 0$ , given a value for  $\xi$  and a realization of the state of the economy  $s_i$ , equations (31) and (33)-(35) from a static system that can be solved for the Ramsey allocation  $c(\xi, s_i)$ ,  $l(\xi, s_i)$ ,  $v(\xi, s_i)$ , and  $\psi(\xi, s_i)$ .
- For  $t = 0$ , given a value for  $\xi$  and an initial state,  $s_0 = (g_0, z_0)$ , the variables  $c_0$ ,  $l_0$ ,  $v_0$ , and  $\psi_0$  are the solution to (31) and (33)-(35) with  $\mathbb{I}(t = 0) = 1$ .

## Quantitative Solution: Step 5

- **Step 5:** Now check whether the guess of  $\xi$  is the correct one in the Ramsey equilibrium, by evaluating the implementability constraint, equation (32).
- Compute the LHS and the RHS, respectively,

$$\begin{aligned}\text{LHS}(\xi, s_0) &= \text{lhs}_0(\xi, s_0) + \beta \Phi(s_0)(I - \beta \Phi)^{-1} \text{lhs}(\xi), \\ \text{RHS}(\xi, s_0) &= \frac{U_c(c_0(\xi, s_0), h_0(\xi, s_0))}{\gamma(\nu_0(\xi, s_0))} d_0,\end{aligned}$$

where  $\text{lhs}(\xi) = [\text{lhs}(\xi, s_i)]$ .

- Then compute the difference

$$y(\xi, s_0) = \text{LHS}(\xi, s_0) - \text{RHS}(\xi, s_0),$$

and use a numerical solver to find  $\xi$  such that  $y(\xi, s_0) = 0$ .

- This yields the equilibrium value of  $\xi$  (and with it the equilibrium allocations) for a given initial state  $s_0$ .

# Quantitative Solution: Step 6

- **Step 6:** Finally, we discuss the computation of  $\pi_t$ .
- Because  $p_0$  is assumed to be given, so is  $\pi_0 = P_0/P_{-1}$ .
- Define the real value of debt as  $d_t \equiv (R_{t-1}B_{t-1} + M_{t-1})/P_t$ . Note that  $d_0$  is given.
- Use the sequential government budget constraint to write

$$\pi_{t+1} = R_t/d_{t+1} [d_t + g_t - \tau_t w_t l_t - c_t/v_t(1 - R_t^{-1})], \quad t \geq 0.$$

- We can prove that  $d_{t+1} = [\gamma(v_{t+1})/U_{c,t+1}] \text{LHS}(t+1)$ . Then construction of  $\text{LHS}(t)$  is similar to that of  $\text{LHS}(\xi, s_0)$  described above.
- Indeed, given  $\xi$ ,  $\text{LHS}(t)$  ( $t > 0$ ) depends only on  $s_t$ .

## Further Question

- How about the economy with price stickiness?
- One way: as Aiyagari, Marcet, Sargent, and Seppälä (2002), to prove the time consistency.
- Another way: as Marcet and Marimon (2019), to rewrite the Ramsey problem into a recursive form.

Questions are welcome.